



Kanea Chatbot

Response-Quality Evaluation — ROUGE-1, ROUGE-2 & BERTScore

Scoring the LLM's actual generated text against gold-reference answers — a companion to the main Evaluation Framework's Section 6

Component covered: Full pipeline output text (intent → routing → RAG → Groq LLM generation)

Reproduce with: `python backend/eval/evaluate_response_quality.py`

Test data: `backend/eval/test_response_quality.json` (20 cases)

Result data: `backend/eval/results_response_quality.json`

Date generated: 2 August 2026

Table of Contents

1. Executive Summary
2. Why This Evaluation Exists
3. Methodology
4. Metric Definitions — ROUGE-1, ROUGE-2, BERTScore
5. Results — Aggregate Scores
6. Results — Per-Case Scores
7. Discussion & Findings
8. Limitations of This Evaluation
9. Recommendations
10. Appendix — Full Case Transcripts

1. Executive Summary

This report scores the **actual text the Kanea chatbot generates** — not the routing decisions or retrieval hit-rate that `EVALUATION.md` already covers, but the sentences a customer would actually read. Twenty realistic ECG/PURC customer questions were sent through the **real, live pipeline** (real Groq LLM calls, no mocking) and the replies were compared against hand-written gold-reference answers grounded in the same knowledge base Kanea itself retrieves from, using three complementary automatic metrics: **ROUGE-1**, **ROUGE-2**, and **BERTScore**.

Headline numbers (n = 20 live cases)

Metric	Precision	Recall	F1
ROUGE-1 (unigram overlap)	0.588	0.426	0.474
ROUGE-2 (bigram overlap)	0.330	0.254	0.278
BERTScore (semantic similarity, roberta-large)	0.912	0.886	0.898

Three findings drive the rest of this report:

- **The BERTScore/ROUGE gap is the headline result.** BERTScore F1 (0.898) is far higher than ROUGE-2 F1 (0.278) because Kanea's LLM *paraphrases* the knowledge base rather than copying it verbatim — exactly what a helpful assistant should do, and exactly what n-gram metrics are least equipped to reward. See §7.1.
- **3 of 20 live calls (15%) hit a Groq API error** and fell back to Kanea's built-in "I'm having a technical issue" message instead of a real answer — a reliability finding, not a language-quality one. See §7.2.
- **7 of 20 static, FAQ-style questions (35%) were mis-routed to the "dynamic" (live outage) path** by the keyword intent classifier, reproducing — with real evaluation queries — the exact keyword-tie weakness [\[EVALUATION.md\]](#) §7.2 already flagged as deferred future work. This measurably cost quality: mis-routed cases scored ~3 BERTScore points and ~11 ROUGE-1 points lower on average. See §7.3.

2. Why This Evaluation Exists

Kanea's existing evaluation framework ([EVALUATION.md](#) , [backend/eval/evaluate.py](#)) deliberately scores five separate pipeline layers — intent classification, query routing, the safety-critical danger trigger, RAG retrieval, and live operational metrics — because a single pipeline-wide "accuracy" number can't describe a system built from that many independent decision points. That framework explicitly named one layer it could not score automatically:

From EVALUATION.md §6 — "Response-quality (qualitative, not automated here)"

"The LLM's actual generated text (tone, faithfulness to the retrieved context, conciseness) isn't scored by evaluate.py — doing that automatically needs either an LLM-as-judge pass or a human rubric, and both cost real API calls / time that felt out of scope for an offline, free-to-re-run script."

This report fills that specific gap using the standard NLP toolkit for scoring generated text against a reference: **ROUGE** (lexical n-gram overlap, the standard for summarization/QA evaluation since 2004) and **BERTScore** (contextual-embedding similarity, 2019, designed specifically to catch paraphrases that ROUGE penalizes). It does not replace the human-rubric approach §6 also recommends for tone/faithfulness judgement — see §8 for where the two approaches differ and why both still have a place.

3. Methodology

3.1 Test set construction

20 customer questions were written to span every static, RAG-answerable section of Kanea's knowledge base — 14 sections from `ecg_knowledge.txt` and 6 from `purc_knowledge.txt` — deliberately mirroring the section coverage of the existing RAG retrieval test set (`backend/eval/test_retrieval.json`) so the two evaluations can be read side by side. For each question a **gold-reference answer** was hand-written, grounded strictly in the same source text Kanea's retriever can see — never containing a fact the knowledge base doesn't state.

Dynamic queries (live outage status) and escalation queries were intentionally excluded from this particular test set: their "correct" answer depends on mutable live-scraped data or on a fixed hand-off script, not on stable reference text, so ROUGE/BERTScore against a static gold answer would not be a meaningful comparison for them. Static, knowledge-grounded FAQ questions are where text-similarity-to-a-reference is actually the right evaluation tool.

3.2 What was actually run

Each of the 20 queries was sent through `chatbot.get_bot_response()` — the exact function `main.py` calls for a real user message — with a fresh session per case. This exercises the **entire live pipeline**: keyword intent classification → query-type routing → TF-IDF RAG retrieval over the ECG/PURC knowledge base → system-prompt construction → a real network call to the Groq API (`llama-3.3-70b-versatile`), temperature 0.4) → escalation-tag stripping. No component was mocked or stubbed; the candidate answers scored below are exactly what a real Kanea user would have received for that message.

3.3 Scoring

Each (reference, candidate) pair was scored with:

- `rouge-score` v0.1.2 — Google's reference implementation, `use_stemmer=True` , for ROUGE-1 and ROUGE-2.
- `bert-score` v0.3.13 — the reference implementation from Zhang et al. (2020), model `roberta-large` (the package's own English default), `lang="en"` , no baseline rescaling.

Both are open-source, deterministic-given-the-same-text libraries; the only source of run-to-run variance is the Groq LLM's own sampling temperature, not the scoring itself.

4. Metric Definitions — ROUGE-1, ROUGE-2, BERTScore

4.1 ROUGE-1 and ROUGE-2 (lexical n-gram overlap)

ROUGE-N measures how many contiguous sequences of N words ("n-grams") the candidate text shares with the reference text. **ROUGE-1** compares single words (unigrams); **ROUGE-2** compares consecutive word pairs (bigrams), which makes it far stricter — it only rewards overlap when the candidate uses the reference's exact word *order*, not just its vocabulary.

```
Precision = (matching n-grams) / (n-grams in candidate)
Recall    = (matching n-grams) / (n-grams in reference)
F1        = 2 * Precision * Recall / (Precision + Recall)
```

Precision asks "how much of what the bot said is grounded in the reference wording"; Recall asks "how much of the reference's content did the bot's answer cover." `use_stemmer=True` means "connection" and "connections" count as the same token, so the score isn't punished by routine morphology.

4.2 BERTScore (contextual semantic similarity)

BERTScore replaces exact string matching with **meaning** matching. Both the candidate and reference are passed through a pretrained transformer (here, `roberta-large`) to get a contextual embedding vector for every token — the same word gets a different vector depending on its sentence, which is what lets it recognize that "reconnect your meter" and "restore your electricity supply" are saying almost the same thing even though they share zero words.

```
For each candidate token: find its highest-cosine-similarity match among reference tokens → avg = Precision
For each reference token: find its highest-cosine-similarity match among candidate tokens → avg = Recall
F1 = 2 * Precision * Recall / (Precision + Recall)
```

This greedy, order-independent matching is exactly why BERTScore tolerates paraphrasing and reordering that ROUGE-2 penalizes heavily — and exactly why it must be read *alongside* ROUGE, not instead of it: BERTScore can score two sentences as highly similar even when they diverge on a specific fact (a date, a phone number, a tariff figure), because a single wrong token barely moves an average over dozens of embedding dimensions. §7.1 and §7.4 show concrete cases of this trade-off in Kanea's own results.

4.3 Why all three, together

Metric	Rewards	Blind to
ROUGE-1	Using the reference's vocabulary	Word order; synonyms; paraphrase
ROUGE-2	Using the reference's exact phrasing	Any valid rewording — heavily penalizes a fluent paraphrase
BERTScore	Preserving meaning, in any wording	Specific facts/numbers being swapped or dropped

No single one of these three numbers can answer "is Kanea's response good?" on its own — that is precisely why `EVALUATION.md`'s framework already scores the pipeline in layers rather than with one number, and why this report keeps that same principle for the generation layer.

5. Results — Aggregate Scores

Three of the twenty live calls returned Kanea's built-in Groq-error fallback text instead of a real generated answer (see §7.2) — a pipeline-reliability event, not a language-generation one. Reporting *only* the all-20 average would understate how well the LLM writes when the API actually responds; reporting *only* the 17-case average would hide a real reliability issue a user would experience. Both are shown.

■ All 20 cases (as observed, incl. API failures) ■ 17 cases (excluding the 3 API failures)

ROUGE-1 F1



ROUGE-2 F1



BERTScore F1



Scope	ROUGE-1 P	ROUGE-1 R	ROUGE-1 F1	ROUGE-2 P	ROUGE-2 R	ROUGE-2 F1	BERTScore P	BERTScore R	BERTScore F1
All 20 cases	0.588	0.426	0.474	0.330	0.254	0.278	0.912	0.886	0.898
17 cases (API failures excluded)	0.661	0.493	0.545	0.389	0.299	0.328	0.928	0.900	0.914

How to read a BERTScore F1 around 0.90–0.91

BERTScore F1 is not scaled 0–1 the way accuracy is — even two *unrelated* English sentences typically score 0.80–0.85 on `roberta-large`, because most of the score reflects shared function words, grammar, and topic-level embedding similarity rather than content overlap. A score in the 0.90s indicates strong content alignment for this model; it should be read relative to the per-case range in §6 (0.797–0.992 here), not against an absolute 0–1 scale.

6. Results — Per-Case Scores

All 20 cases, sorted by ID. Bars are drawn on the same 0–1 scale across all three metrics so the ROUGE-2 vs. BERTScore gap discussed in §7.1 is directly visible row by row.

■ ROUGE-1 F1 ■ ROUGE-2 F1 ■ BERTScore F1

ID	Query	Route	ROUGE-1 F1	ROUGE-2 F1	BERTScore F1
q-1	What is ECG and when was it established?	static	 0.206	 0.057	 0.843
q-2	Which districts are part of the Tema region?	static	 0.971	 0.909	 0.992
q-3	Why do I still have to pay a charge on my bill even though I don't use much electricity?	static	 0.429	 0.159	 0.889
q-4	My prepaid meter card got lost, what should I do?	static	 0.658	 0.394	 0.938
q-5	What documents do I need to apply for a new low voltage electricity connection?	dynamic†	 0.600	 0.392	 0.921
q-6	How do I get my power reconnected after it was disconnected for non-payment?	static	 0.707	 0.525	 0.952
q-7	What counts as an illegal electricity connection?	dynamic†	 0.588	 0.280	 0.918
q-8	What should I do if I see a fallen or sagging power line?	dynamic†	 0.500	 0.265	 0.917
q-9	How do I report a fault to ECG?	static	 0.761	 0.638	 0.957
q-10	What usually causes unplanned power outages in Ghana?	dynamic†	 0.484	 0.280	 0.910
q-11	How can I apply to get street lighting installed in my area?	static	 0.654	 0.324	 0.928
q-12	What is PURC and how is it different from ECG?	static	 0.596	 0.282	 0.917
q-13	What is ECG's main customer service phone number?	static	 0.275	 0.082	 0.900
q-14	What is the current electricity tariff for a residential customer using 200 units a month?	dynamic†	 0.130	 0.019	 0.817
q-15	What is the Lifeline electricity tariff band and who does it apply to?	static	 0.636	 0.438	 0.940
q-16	What tariff applies to a small business or shop?	static, API failed‡	 0.077	 0.000	 0.803
q-17	How are large industrial customers billed for electricity?	dynamic†	 0.485	 0.277	 0.892
q-18	How does billing work for a customer with rooftop solar under net metering?	static, API failed‡	 0.065	 0.000	 0.797
q-19	Why do ECG's electricity tariffs change so often?	dynamic†	 0.581	 0.246	 0.901
q-20	If ECG doesn't resolve my billing complaint, what can I do?	static, API failed‡	 0.082	 0.000	 0.835

† "dynamic" = the query was actually static/FAQ in nature but the keyword classifier routed it to the dynamic (live-outage) path — see §7.3. ‡

"API failed" = Groq call raised inside `chatbot.py`'s own try/except and returned the built-in fallback message — see §7.2.

7. Discussion & Findings

7.1 The BERTScore/ROUGE-2 gap is the paraphrasing signature, not a quality problem

Averaged across all 20 cases, ROUGE-2 F1 is 0.278 while BERTScore F1 is 0.898 — a 62-point gap on a comparable 0–1 scale. Reading the actual transcripts (Appendix, §10) shows why: Kanea's LLM almost never copies the reference's exact phrasing. For q-9 ("How do I report a fault to ECG?"), the reference reads *"you can report a fault by contacting your ECG District Office, calling 0302 611 611..."*; the candidate reads *"To report a fault, you can contact your District Office, call 0302 611 611..."* — the same facts, reordered and reworded, which ROUGE-2 scores relatively well here (0.638) precisely because the phrase structure happens to survive. Cases like q-1 and q-13 show the more typical pattern: correct content, restructured sentence, ROUGE-2 near zero, BERTScore still 0.84–0.90. This is exactly the failure mode BERTScore's design paper identifies ROUGE-N as vulnerable to, and it is the concrete justification — visible in Kanea's own output, not just in the abstract — for reporting BERTScore alongside ROUGE rather than ROUGE alone.

The reverse caution matters equally: BERTScore does **not** verify facts. q-1 ("What is ECG and when was it established?") scores a respectable 0.843 BERTScore F1 even though the candidate answer declines to give a founding date at all (see §7.4) — the response is topically on-target and fluently written, which is most of what BERTScore measures, but it doesn't answer the question asked. A high BERTScore is evidence of fluent, on-topic, semantically-aligned text; it is not, by itself, evidence of factual completeness.

7.2 Reliability: 3 of 20 live calls (15%) hit an internal error fallback

q-16, q-18, and q-20 all returned the identical string: *"I'm having a technical issue right now. Please call 0800-POWER (0800-76937) for immediate assistance."* — the hard-coded fallback inside `chatbot.py`'s `try/except` around the Groq HTTP call (`backend/chatbot.py`, `get_bot_response`). Because that exception is caught and replaced with fallback text rather than propagated, the specific cause (timeout, rate limit, or transient network error) isn't visible from outside the function — this evaluation ran 20 requests in quick succession with only a 0.5s gap between them, which is a plausible trigger. These three cases pull every aggregate metric down substantially (their BERTScore F1 sits at 0.797–0.835, noticeably below the 0.898 all-case mean) purely because the fallback sentence is semantically unrelated to the reference answer — not because the LLM generated a poor answer to the question. This is why §5 reports both the all-20 and the 17-excluding-failures numbers side by side: the 17-case figures (ROUGE-1 F1 0.545, BERTScore F1 0.914) are the fairer read of *generation quality*; the all-20 figures are the fairer read of what a real user actually experiences, reliability included.

7.3 Intent-routing errors measurably reduce answer quality

7 of the 20 static, FAQ-style questions (35%) were classified as `query_type = "dynamic"` by `chatbot.classify_intent()` / `classify_query_type()` — the path meant for live outage questions, not billing or safety FAQs. This is not a new bug: [EVALUATION.md](#) §1 already documented that the intent classifier is pure keyword substring matching with no tie-breaking logic, and §7.2 named "replace the keyword classifier" as deferred future work. What this evaluation adds is **a measured, real-query cost of leaving that gap open**:

Group	n	Avg RAG docs retrieved	Avg ROUGE-1 F1	Avg BERTScore F1
Correctly routed to <code>static</code>	10	3.0	0.589	0.926
Mis-routed to <code>dynamic</code>	7	2.0	0.481	0.896

(Excludes the 3 API-failure cases from §7.2 so routing effects aren't confounded with them.) The mechanism is visible in the code: `_build_rag_context()` (`backend/chatbot.py`) fetches `top_n=5` documents for a `static` query but only `top_n=2` for a `dynamic` one, on the assumption that dynamic queries need RAG only as light "supplemental background." A mis-routed FAQ question therefore gets less than half the knowledge-base context it should, which shows up here as a real, if modest, drop in both metrics — larger in ROUGE-1 (–11 points) than in BERTScore (–3 points), consistent with BERTScore's greater tolerance for a shorter-but-still-on-topic answer. The specific trigger is the same one [EVALUATION.md](#) already diagnosed: several keywords appear in more than one intent's keyword list (e.g. "electricity" and "voltage" under `outage_status`, "connection" and "apply" under `billing`), `classify_intent()` breaks ties with Python's `max()` over a dict, and `outage_status` is declared first in

`INTENT_KEYWORDS` — so every tie defaults to `outage_status`. All 7 mis-routed cases here are consistent with that exact tie pattern.

7.4 RAG retrieval misses cause honest — but unhelpful — refusals

q-1 ("What is ECG and when was it established?") retrieved `purc_overview` and `ecg_prepaid_meters` instead of the correct `ecg_company_overview` section, and the LLM correctly declined to invent a founding date it wasn't given — matching the system prompt's explicit instruction not to speculate. q-14 ("current electricity tariff for a residential customer using 200 units a month") retrieved `purc_non_residential_electricity_tariffs` and `purc_special_load_tariffs` instead of the correct `purc_residential_electricity_tariffs` section, and the candidate answer states *"I'm not provided with the current residential tariff information"* rather than quoting a non-residential rate as if it applied to a residential customer. In both cases this is the safer of two failure modes — Kanea did not hallucinate a specific fact, which is the behavior `EVALUATION.md`'s response rules are designed to enforce — but it is still a usefulness failure traceable to the TF-IDF retriever: q-1's query uses "established" where the source text says "commenced," and q-14's "200 units a month" doesn't share literal tokens with "0-300 kWh," so neither query's bag-of-words overlaps enough with the correct section to clear `MIN_RELEVANCE_SCORE` ahead of a competing section. This is the same literal-token-overlap limitation `EVALUATION.md` §4 already measured (Precision@5 = 0.463) and §7.2 named as the reason a move to semantic/embedding-based retrieval was deferred rather than attempted — this report's q-1 and q-14 are concrete real-question instances of exactly that gap.

7.5 Best and weakest cases

Best: q-2 — "Which districts are part of the Tema region?" (ROUGE-1 F1 0.971, BERTScore F1 0.992)

A closed-list factual question with one unambiguous correct source document. The candidate reproduced the district list almost verbatim, which is exactly the kind of question where high lexical overlap is both achievable and a legitimate quality signal — the "right" answer has very little room for valid paraphrase.

Weakest (excluding API failures): q-14 — tariff lookup with a RAG miss (ROUGE-1 F1 0.130, BERTScore F1 0.817)

Covered in §7.4 above — a retrieval failure, not a generation failure. The lowest scores in this dataset trace back to retrieval or infrastructure, not to the LLM writing a poor sentence once it has the right facts in front of it.

8. Limitations of This Evaluation

- **n = 20 is a small sample.** Individual case scores are illustrative; the averages are indicative, not statistically definitive. The set was sized to give complete, hand-verifiable coverage of every static knowledge-base section rather than to support tight confidence intervals.
- **One reference answer per query.** ROUGE and BERTScore both compare against a single gold answer; a differently-phrased but equally correct response can legitimately score lower. Multi-reference scoring (several acceptable gold answers per query) would tighten this but multiplies the hand-authoring effort.
- **Non-deterministic generation.** Groq calls used temperature 0.4, so re-running this exact script will not reproduce identical scores — the API-failure count and exact per-case numbers are a snapshot of one run, though the qualitative findings (§7.1–§7.4) reflect structural properties of the pipeline, not one-off luck.
- **BERTScore's model is general-purpose, not domain-tuned.** `roberta-large` was pretrained on general English text, not ECG/utility terminology specifically; it has no special notion of a correct tariff figure or phone number — see §7.1's caution about facts.
- **Scope is static/RAG-answerable queries only.** Dynamic (live outage) and escalation-path responses are not covered by this metric pair, for the reason given in §3.1 — a live-data or fixed-script answer isn't well served by comparison to a static gold reference.
- **This complements, not replaces,** the human-scored faithfulness/tone/relevance rubric `EVALUATION.md` §6 recommends. ROUGE and BERTScore can tell you a response is lexically or semantically close to a reference; they cannot tell you it is empathetic, correctly toned for a utility customer, or appropriately concise — that still needs a human rater.

9. Recommendations

#	Recommendation	Why (tied to this evaluation)
1	Add retry-with-backoff around the Groq HTTP call in <code>get_bot_response()</code> before falling back to the generic error message.	Would likely have recovered some or all of the 3/20 (15%) API-failure cases in §7.2.
2	Give <code>classify_intent()</code> an explicit tie-break rule (e.g. score margin, or reordering <code>INTENT_KEYWORDS</code> so <code>outage_status</code> isn't the default tie-winner) instead of relying on dict-iteration order.	Directly targets the mechanism behind the 7/20 (35%) static→dynamic mis-routing measured in §7.3.
3	When a <code>dynamic</code> -routed query returns zero matching live outages, consider retrieving at the <code>static</code> <code>top_n=5</code> rather than <code>top_n=2</code> , since the query is evidently not actually about a live incident.	Would recover most of the ~3-point BERTScore / ~11-point ROUGE-1 gap in §7.3 even without fixing the underlying misclassification.
4	Re-run <code>evaluate_response_quality.py</code> whenever <code>chatbot.py</code> , <code>knowledge_base.py</code> , or the knowledge-base text files change, alongside the existing <code>evaluate.py</code> regression check.	Keeps this a living regression check, the same role <code>EVALUATION.md</code> already assigns its own test sets.
5	Add 2–3 more gold references per query where a genuinely valid paraphrase exists (multi-reference ROUGE/BERTScore).	Directly addresses the single-reference limitation in §8 without requiring a larger query set.

10. Appendix — Full Case Transcripts

Every query, its gold-reference answer, and the actual live candidate answer Kanea generated, for full transparency and manual spot-checking.

q-1 · *"What is ECG and when was it established?"*

REFERENCE

ECG (Electricity Company of Ghana) is a limited liability company wholly owned by the Government of Ghana. Ghana's first government-sponsored public electricity supply began in 1914, and ECG remains the primary electricity distributor across southern Ghana, serving residential, commercial, and industrial customers. Its mission is to provide quality, reliable, and safe electricity services to support Ghana's economic growth, guided by the ENERGY core values: Excellence, Nurturing, Ethical, Reliability, Growth, and You.

CANDIDATE

ECG stands for Electricity Company of Ghana. However, the provided knowledge base does not specify when ECG was established. For more information, you can contact ECG's general support at 0302 611 611.

Intent: general · Route: static · RAG sources: purc_overview, ecg_prepaid_meters (miss — see §7.4) · ROUGE-1 F1 0.206 · ROUGE-2 F1 0.057 · BERTScore F1 0.843

q-2 · *"Which districts are part of the Tema region?"*

REFERENCE

The Tema Region covers districts including Ada, Afienya, Ashaiman, Juapong, Krobo, Nungua, Prampram, Tema North, and Tema South.

CANDIDATE

The Tema Region districts include: Ada, Afienya, Ashaiman, Juapong, Krobo, Nungua, Prampram, Tema North, and Tema South.

Intent: general · Route: static · RAG sources: ecg_service_areas, ecg_contact_directory · ROUGE-1 F1 0.971 · ROUGE-2 F1 0.909 · BERTScore F1 0.992

q-3 · *"Why do I still have to pay a charge on my bill even though I don't use much electricity?"*

REFERENCE

In addition to charges for the electricity you actually consume, ECG bills include a fixed monthly service charge that covers maintenance of the network equipment serving your connection. This service charge applies every month regardless of how much power you use, and it continues to accrue even if your meter has been disconnected, since your account remains active in ECG's database.

CANDIDATE

The charge you're referring to is likely the service charge. According to our billing information, this is a monthly charge for customer services rendered, covering maintenance of the network equipment, and it continues even when a meter is disconnected. If you're a low-consumption household, you might be eligible for the Lifeline band, which has a lower service charge. You can check the PURC website or visit an ECG office for more information on residential tariffs.

Intent: billing · Route: static · RAG sources: 5 docs retrieved · ROUGE-1 F1 0.429 · ROUGE-2 F1 0.159 · BERTScore F1 0.889

q-4 · *"My prepaid meter card got lost, what should I do?"*

REFERENCE

If your prepaid card is missing, report it at your ECG District Office along with a copy of your vending receipt. ECG will issue you a new programmed card, and a replacement fee applies.

CANDIDATE

To replace a lost prepaid meter card, register at your District Office with a copy of your vending receipt. They will issue a new programmed card for a replacement fee. Contact your District Office for more information and assistance.

Intent: billing · Route: static · RAG sources: ecg_prepaid_meters · ROUGE-1 F1 0.658 · ROUGE-2 F1 0.394 · BERTScore F1 0.938

q-5 · "What documents do I need to apply for a new low voltage electricity connection?"

REFERENCE

For a new low voltage connection, submit a completed service form with a GHC 10 registration fee, together with an Installation Completion Certificate signed by a licensed electrician, a valid ID (National ID, Passport, Voter's ID, or Driver's License), and the original plus a photocopy of your site plan. Businesses must also provide proof of business operation. After an ECG inspection and payment of the cost estimate, connection is completed within 3 working days if lines already exist, or up to 5 days for extensions of up to 2 poles.

CANDIDATE

To apply for a new low voltage electricity connection, you'll need to submit a completed service form, site plan, and ID. Additionally, you'll need an Installation Completion Certificate signed by a licensed electrician, and proof of identification such as a National ID, Passport, Voters ID, or Drivers License. Businesses must also provide proof of business operation.

Intent: outage_status (mis-classified — see §7.3) · Route: dynamic · RAG sources: ecg_new_connection, ecg_safety_guidelines · ROUGE-1 F1 0.600 · ROUGE-2 F1 0.392 · BERTScore F1 0.921

q-6 · "How do I get my power reconnected after it was disconnected for non-payment?"

REFERENCE

To get reconnected, settle all outstanding bills and pay the approved reconnection fee at your ECG District Office. Your meter will then be reconnected by ECG's reconnection team, usually the same day, once a team is available.

CANDIDATE

To reconnect after disconnection, settle all outstanding bills and pay the approved reconnection fee at your District Office. Your meter will be reconnected as soon as the reconnection team is available, usually within the same day. Please contact your District Office for more information.

Intent: billing · Route: static · RAG sources: 5 docs retrieved · ROUGE-1 F1 0.707 · ROUGE-2 F1 0.525 · BERTScore F1 0.952

q-7 · "What counts as an illegal electricity connection?"

REFERENCE

An illegal connection is obtaining electricity from an unauthorized source, which includes direct connections without a meter, unapproved meter transfers, meter tampering or bypass, unauthorized network extensions, unauthorized redistribution of supply, or reconnecting your own supply after a legal disconnection. Only qualified, authorized ECG personnel may reconnect a meter, and there is a cash reward for reporting confirmed power theft to 0302 611 611.

CANDIDATE

An illegal connection is the acquisition of electricity services from an unauthorized person or source. Examples include: direct connections without meters, meter transfers without approval, meter tampering, meter bypass, unauthorized network extensions, or self-reconnection after legal disconnection.

Intent: outage_status (mis-classified — see §7.3) · Route: dynamic · RAG sources: ecg_disconnection_reconnection, ecg_illegal_connections · ROUGE-1 F1 0.588 · ROUGE-2 F1 0.280 · BERTScore F1 0.918

q-8 · "What should I do if I see a fallen or sagging power line?"

REFERENCE

Stay well away from a sagging or fallen power line. Do not touch the wire or anything it is in contact with, such as metal fences or standing water, and do not attempt to move it, since even a fallen tree touching a line can conduct high voltage. Call ECG immediately on 0302 611 611, or the emergency line 0800-POWER (0800-76937), and do not approach the area.

CANDIDATE

If you see a fallen or sagging power line, stay away at a distance. Do not touch the wire or contact objects it touches. Call ECG immediately on 0302 611 611.

Intent: outage_status (mis-classified — see §7.3) · Route: dynamic · RAG sources: ecg_safety_guidelines, ecg_prepaid_meters · ROUGE-1 F1 0.500 · ROUGE-2 F1 0.265 · BERTScore F1 0.917

q-9 · "How do I report a fault to ECG?"

REFERENCE

You can report a fault by contacting your ECG District Office, calling 0302 611 611 or 0551 444 011, or using the online portal at cx.eservicesafrica.com. Emergency faults are prioritized, and unresolved complaints can be escalated to PURC at purc.com.gh.

CANDIDATE

To report a fault, you can contact your District Office, call 0302 611 611 or 0551 444 011, or use the online portal at cx.eservicesafrica.com.

Intent: fault_report · Route: static · RAG sources: 4 docs retrieved · ROUGE-1 F1 0.761 · ROUGE-2 F1 0.638 · BERTScore F1 0.957

q-10 · *"What usually causes unplanned power outages in Ghana?"*

REFERENCE

Unplanned outages are commonly caused by broken high-tension poles from vehicle accidents or weather, faults on 11kV or 33kV feeder lines, transformer overloads or failures, underground or overhead cable faults, fire outbreaks near equipment, and upstream challenges from GRIDCo or VRA. Restoration time depends on the fault type: broken poles may take several hours to fix, while transformer faults can take a day or more.

CANDIDATE

Unplanned power outages in Ghana are commonly caused by factors such as broken high-tension poles, feeder faults, transformer overloads or failures, cable faults, and fire outbreaks near equipment.

Intent: outage_status (topic-correct here) · Route: dynamic · RAG sources: ecg_outage_causes, ecg_recent_news · ROUGE-1 F1 0.484 · ROUGE-2 F1 0.280 · BERTScore F1 0.910

q-11 · *"How can I apply to get street lighting installed in my area?"*

REFERENCE

To install street lighting, apply through your local Assembly or ECG District Manager with a layout of the area. ECG invites estimates from certified contractors, the General Manager verifies transformer capacity, the applicant pays a deposit, and ECG then authorizes a contractor to proceed. The contractor is paid from the deposited amount once the work is completed.

CANDIDATE

To apply for street lighting installation, submit an application through your Assembly or District Manager with an area layout. ECG will then invite estimates from certified contractors, verify transformer capacity, and authorize a contractor to proceed after you pay a deposit. For more information, contact ECG at 0302 611 611.

Intent: billing · Route: static · RAG sources: 3 docs retrieved · ROUGE-1 F1 0.654 · ROUGE-2 F1 0.324 · BERTScore F1 0.928

q-12 · *"What is PURC and how is it different from ECG?"*

REFERENCE

PURC (the Public Utilities Regulatory Commission) is the independent government regulator of electricity and water utilities in Ghana, established in October 1997 under Act 538. PURC is not ECG. It is the body that sets and approves the tariffs ECG is allowed to charge, monitors service standards, and handles complaints when a utility does not resolve a customer's issue, including billing disputes escalated from ECG's District level.

CANDIDATE

PURC (Public Utilities Regulatory Commission) is an independent regulator of electricity and water utility services in Ghana. It sets guidelines for utility rates, examines and approves tariffs, and protects consumer interests. ECG (Electricity Company of Ghana) is a utility company that provides electricity services. PURC is not ECG, but rather the independent body that oversees and regulates ECG's activities, including setting and approving the rates ECG can charge. For complaints or issues not resolved by ECG, customers can escalate to PURC.

Intent: general · Route: static · RAG sources: purc_overview, purc_functions, purc_tariff_regulatory_context · ROUGE-1 F1 0.596 · ROUGE-2 F1 0.282 · BERTScore F1 0.917

q-13 · *"What is ECG's main customer service phone number?"*

REFERENCE

ECG's main call centre can be reached on 0302 611 611 or 0551 444 011, available 24 hours for fault reporting and emergencies. For electrical hazards or fallen wires, call the dedicated emergency line 0800-POWER (0800-76937).

CANDIDATE

The main customer service phone number for ECG is 0302 611 611.

Intent: general · Route: static · RAG sources: none retrieved (query below TF-IDF relevance floor) · ROUGE-1 F1 0.275 · ROUGE-2 F1 0.082 · BERTScore F1 0.900

q-14 · *"What is the current electricity tariff for a residential customer using 200 units a month?"*

REFERENCE

For a residential customer using 200 kWh a month, that usage falls in the 0-300 kWh band, billed at 203.7509 GHp/kWh, plus the standard residential service charge, effective 1st July 2026. Customers in the Lifeline band (0-30 kWh) pay a lower rate of 89.9315 GHp/kWh with a reduced Lifeline service charge. Since PURC revises tariffs roughly every quarter, confirm the current rate via the ECG PowerApp or purc.com.gh.

CANDIDATE

I'm not provided with the current residential tariff information. However, I can help with outage-related queries or direct you to our general support line for further assistance: 0302 611 611.

Intent: outage_status (mis-classified — see §7.3) · Route: dynamic · RAG sources: purc_non_residential_electricity_tariffs, purc_special_load_tariffs (miss — see §7.4) · ROUGE-1 F1 0.130 · ROUGE-2 F1 0.019 · BERTScore F1 0.817

q-15 · "What is the Lifeline electricity tariff band and who does it apply to?"

REFERENCE

The Lifeline band covers residential customers using 0-30 kWh per month and exists to keep electricity affordable for the smallest consumers. As of 1st July 2026, it is billed at 89.9315 GHp/kWh with a lower Lifeline service charge of 213.0000 GHp/month, separate from the standard residential service charge.

CANDIDATE

The Lifeline electricity tariff band applies to low-consumption residential households, with a usage of 0-30 kWh per month. The tariff rate for this band is 89.9315 GHp/kWh, with a lower Lifeline service charge of 213.0000 GHp/month. This band is designed to keep electricity affordable for small residential consumers.

Intent: billing · Route: static · RAG sources: 5 docs retrieved · ROUGE-1 F1 0.636 · ROUGE-2 F1 0.438 · BERTScore F1 0.940

q-16 · "What tariff applies to a small business or shop?"

REFERENCE

Non-residential customers such as shops and small offices are billed 183.9549 GHp/kWh for the first 300 kWh per month and 224.0395 GHp/kWh for usage above 301 kWh, plus a non-residential service charge of 1242.8245 GHp/month, effective 1st July 2026. Larger businesses with higher power demand are instead billed under the Special Load Tariff.

CANDIDATE

I'm having a technical issue right now. Please call 0800-POWER (0800-76937) for immediate assistance. *(Groq API error fallback — see §7.2)*

Intent: billing · Route: static · RAG sources: 4 docs retrieved (context was available; the LLM call itself failed) · ROUGE-1 F1 0.077 · ROUGE-2 F1 0.000 · BERTScore F1 0.803

q-17 · "How are large industrial customers billed for electricity?"

REFERENCE

Large industrial or high-demand customers, typically factories or premises connected at medium or high voltage, are billed under Special Load Tariff (SLT) categories rather than the standard non-residential band. For example, SLT-LV customers pay 240.2104 GHp/kWh, and SLT-HV customers pay 188.4762 GHp/kWh, each with a service charge of GHS 500 per month, effective 1st July 2026.

CANDIDATE

Large industrial customers are billed under Special Load Tariffs (SLT) categories, which include SLT-LV, SLT-MV, SLT-MV2, and SLT-HV, with varying energy charges and a monthly service charge of 50,000.0000 GHp (GHS 500/month).

Intent: outage_status (mis-classified — see §7.3) · Route: dynamic · RAG sources: purc_special_load_tariffs, purc_non_residential_electricity_tariffs · ROUGE-1 F1 0.485 · ROUGE-2 F1 0.277 · BERTScore F1 0.892

q-18 · "How does billing work for a customer with rooftop solar under net metering?"

REFERENCE

A net metering customer with rooftop solar pays the Import Tariff for power drawn from ECG's grid and receives a credit at a lower Export Tariff for surplus power sent back to the grid. For example, in the 31-300 kWh band, the Import Tariff is 122.2505 GHp/kWh and the Export Tariff is 81.5004 GHp/kWh, effective 1st July 2026. The Export Tariff is intentionally set lower than the Import Tariff.

CANDIDATE

I'm having a technical issue right now. Please call 0800-POWER (0800-76937) for immediate assistance. *(Groq API error fallback — see §7.2)*

Intent: billing · Route: static · RAG sources: purc_net_metering_tariffs, purc_how_tariffs_are_set (context was available; the LLM call itself failed) · ROUGE-1 F1 0.065 · ROUGE-2 F1 0.000 · BERTScore F1 0.797

q-19 · "Why do ECG's electricity tariffs change so often?"

REFERENCE

Electricity tariffs in Ghana are set and approved by PURC, not ECG. Besides periodic full Major Tariff Reviews, PURC applies a quarterly Automatic Adjustment (Indexation) Formula that revises tariffs to reflect changes in exchange rates, inflation, and fuel costs. Each new quarterly publication explicitly revokes and replaces the previous gazette, which is why customers should confirm the current rate for their billing period via ECG or purc.com.gh rather than relying on an undated figure.

CANDIDATE

ECG's electricity tariffs are approved by PURC and change due to periodic Major Tariff Reviews and routine Automatic Adjustments, which reflect changes in exchange rates, inflation, and fuel costs. Tariffs are typically revised every quarter. For the most current rates, please check the PURC or ECG website.

Intent: outage_status (mis-classified — see §7.3) · Route: dynamic · RAG sources: ecg_billing_and_payment, purc_how_tariffs_are_set · ROUGE-1 F1 0.581 · ROUGE-2 F1 0.246 · BERTScore F1 0.901

q-20 · "If ECG doesn't resolve my billing complaint, what can I do?"

REFERENCE

If a billing dispute or service complaint is not resolved by ECG at the District level, you can escalate it to PURC, which has the statutory power to investigate and settle disputes between customers and utility companies. Complaints can be submitted through purc.com.gh or by calling PURC's Accra office on 0302 218 300.

CANDIDATE

I'm having a technical issue right now. Please call 0800-POWER (0800-76937) for immediate assistance. *(Groq API error fallback — see §7.2)*

Intent: complaint · Route: static · RAG sources: 5 docs retrieved (context was available; the LLM call itself failed) · ROUGE-1 F1 0.082 · ROUGE-2 F1 0.000 · BERTScore F1 0.835